The Credentials Arms Race: Understanding Immigrant Over-Education in the U.S. Labor Market

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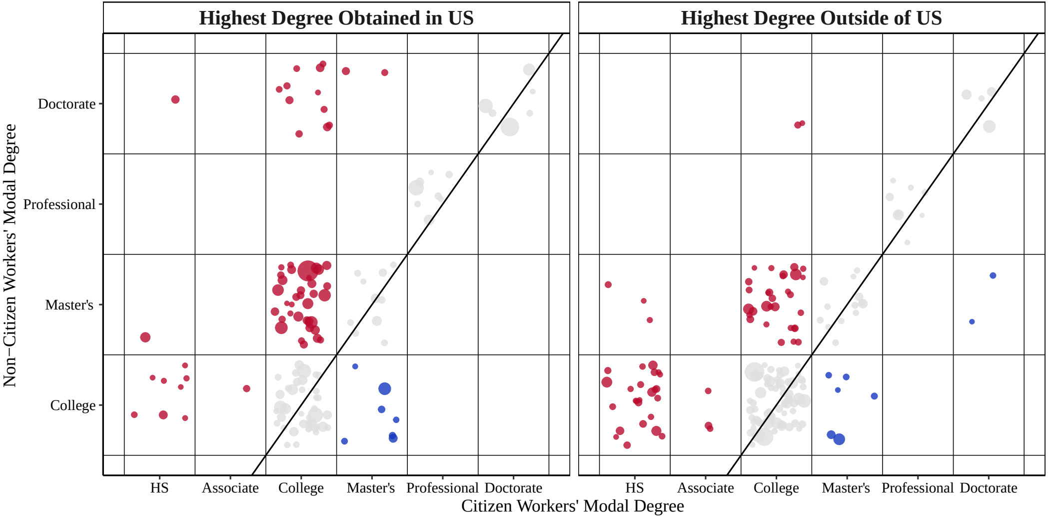
MICHIGAN STATE UNIVERSITY

Motivation

Figure 1: Educational Mismatch of Non-Citizens, ACS 2010-2019



Related Literature

This paper concerns four strands of literature:

1. Over-education and Wage Penalty (Cassidy and Gaulke, 2024; Chiswick et al., 2009, 2013; Hartog, 2000; Kiker et al., 1997; Li et al., 2023; Lu et al., 2021; Verdugo et al., 1989; Warman et al., 2015).

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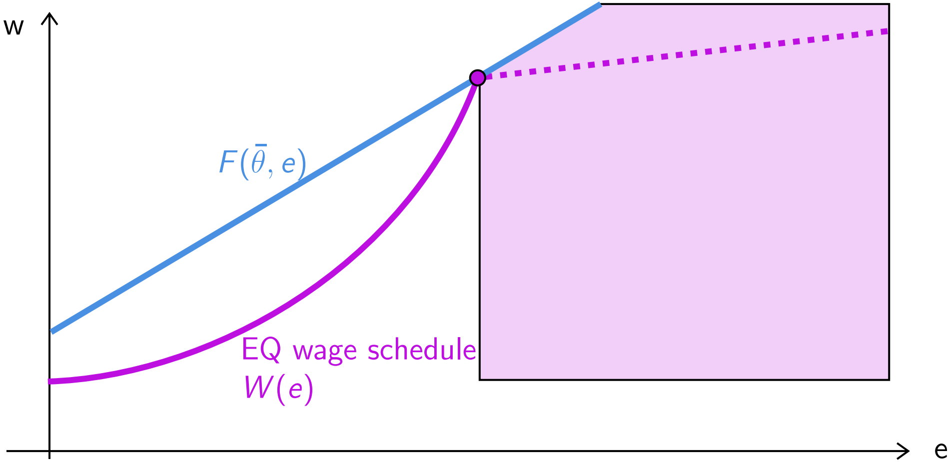
3. Broader Occupational Mismatch (Cassidy, 2019; Chiswick et al., 2013; Hartog, 2000; Imai et al., 2019; Li et al., 2023; Weber et al., 2024).
4. Separating equilibria in games with productive signals (Chatterjee et al., 2021; Cho et al., 1987; Kaymak, 2025; Mailath, 1987; Spence, 1978; Tyler et al., 2000).

Model: Productive Signals with Continuous Types



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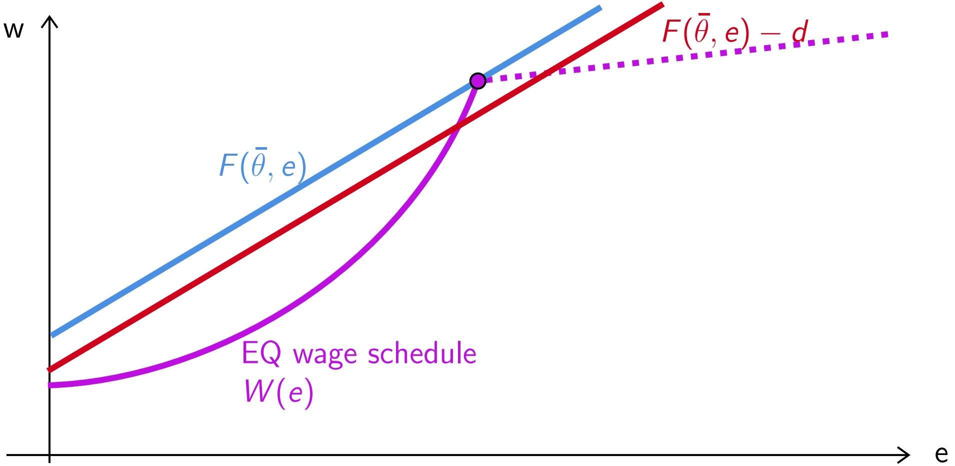

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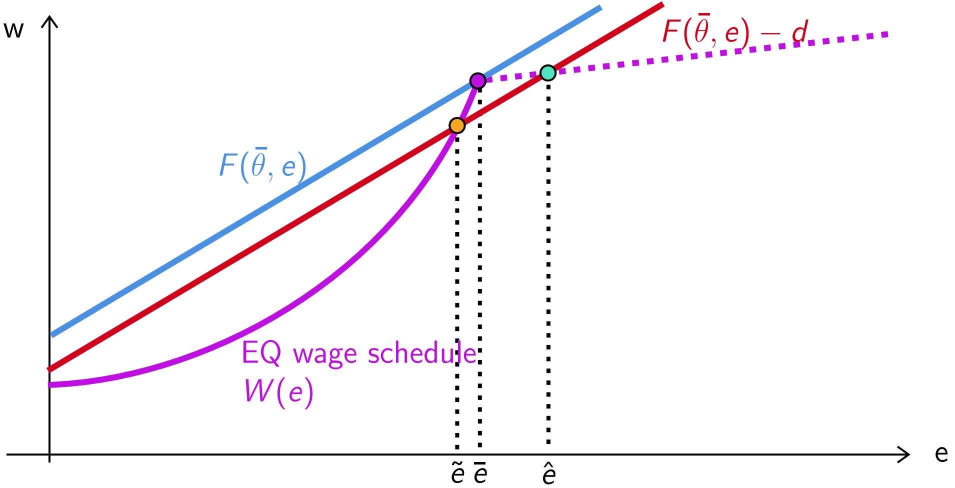

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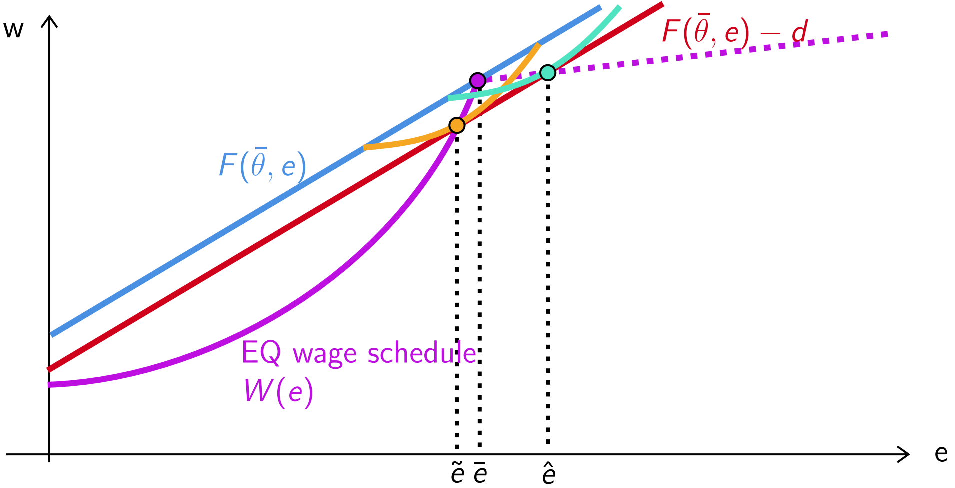

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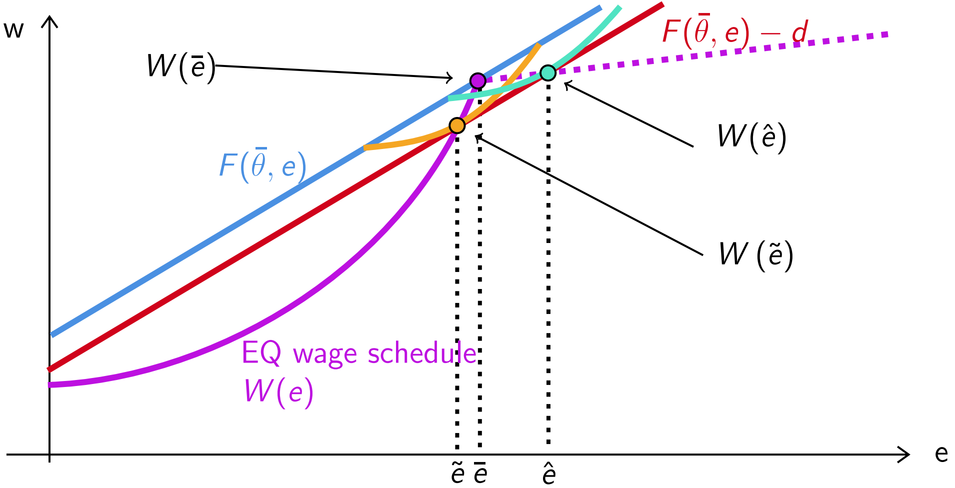

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$$d^s = W(\bar{e}) - \left[E[W(\bar{e} + \alpha)] + \frac{E[W(\bar{e} + \beta)] - E[W(\bar{e} + \alpha)]}{E[\beta - \alpha]} \cdot E[-\alpha] \right].$$

Employer learning implies that $\alpha, \beta \to 0$ over time.

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By comparing within-cohort observed d over time, I can bound d^t and d^s.

Controlling for Job Characteristics

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2. Importance of Certification vs. Typical On-the-Job Training Graphs

LinkedIn Profiles

To study whether employer learning reduces education–occupation gaps—something cross-sectional data (e.g., ACS) cannot address—I construct a panel using scraped LinkedIn profiles from People Data Labs.

Current data include:

- Person record (sex, job, location)
- Job experience (date, company, title, role)
- Education (date, school, degree, major)
- Certifications (date, organization)

Matching LinkedIn Job Titles to SOCs

Using 250 distinct job titles for each of the 1,016 SOC codes from LinkUp, I fine-tune a Llama-3.1 (8B) model to classify each self-reported job title into its corresponding SOC code.

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> *Input: Vice President, company_name*
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Matching LinkedIn Titles to SOCs

Table 1: Example of Predicted SOC Codes using Preliminarily Fine-Tuned Llama-3.1

LinkedIn Title	**O*NET SOC**	**Corresponding O*NET Title**
director - field support	27-2012.05	Media Technical Directors/Managers
senior page designer and copy editor	27-3041.00	Editors
gfliwn askdm	00-0000.00	NA
taylor mattis fellow	51-9195.05	Potters, Manufacturing
solar analyst intern	17-2199.11	Solar Energy Systems Engineers
international visiting attorney	23-1011.00	Lawyers
director of lower school admission	11-9032.00	Education Administrators, Kindergarten through Secondary
asset & reliability engineer	17-2199.00	NA
sales / acct manager	41-1011.00	First-Line Supervisors of Retail Sales Workers
attest intern	13-2099.00	NA
immersion associate	25-1194.00	Career/Technical Education Teachers, Postsecondary
medicaid waiver case manager	11-9151.00	Social and Community Service Managers

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medicaid waiver case manager	11-9151.00	Social and Community Service Managers
網頁工程師	15-1299.08	Computer Systems Engineers/Architects
博士生	25-9044.00	Teaching Assistants, Postsecondary

LinkedIn Panel

I construct a panel of individuals who attended a U.S. degree-granting post-secondary institution and held at least one job matched to an O*NET SOC code. The sample spans 1970–2025 and includes ∼1B person-year observations from ∼53M individuals, averaging 18.6 years per person.

Table 2: Combined Education and Employment Statistics

Part 1: By Highest Degree			**Part 2: Education-to-Employment Gap**		
% Degree	**LinkedIn**	**ACS**	**Gap Duration**	**Count (N)**	**Pct. %**
Assc & Some College	34.8	41.6	Same year (0)	31,339,560	44.6
Bachelor's	36.7	37.5	1 year	14,087,376	20.1
Master's	22.8	15.2	2 years	6,999,362	10.0
Professional	3.2	3.4	3–5 years	10,814,281	15.4
PhD	2.5	2.3	$>$ 5 years	7,015,214	10.0

Constructing Smooth Education Attainment

Using IPEDS data, I can observe variation in the prestige of schools where each worker acquired their degree.

$$\text{score}_s = 0.65 \times \text{Std. SAT/ACT Scores} + 0.35 \times \text{Std. Rejection Rate}$$

At each degree level, I can then apply the calculated score to proxy for the signaling value of the degree.

$$\mathbb{I}_i = \beta_0 + \sum_{d=\text{Mas.}}^{\text{Pro.}} \delta_d \cdot \mathbb{1}\{Deg_i = d\} + \sum_{d=\text{Bac.}}^{\text{Pro.}} \beta_d \cdot \mathbb{1}\{Deg_i = d\} \cdot (score_i - sc\bar{o}re_d) + \varepsilon$$

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$$\hat{e}_i = 1 + \sum_{d=\text{Mas.}}^{\text{Pro.}} \hat{\delta}_d \cdot \mathbb{1}\{Deg_i = d\} + \sum_{d=\text{Bac.}}^{\text{Pro.}} \hat{\beta}_d \cdot \mathbb{1}\{Deg_i = d\} \cdot (score_i - sc\bar{o}re_d)$$

Estimated Education Signals

Table 3: Estimated Marginal Signaling Value of Degrees using Nakao-Treas Prestige Score Graphical Representation

Tier	School	Bachelors	Masters	PhD	Professional
2	Bottom 25%	-2.14	0.96	2.92	9.45
3	Top 75%	-1.92	1.11	3.26	10.18
4	Top 50%	-1.54	1.58	3.52	10.67
5	Top 33%	-1.33	1.54	3.28	11.07
6	Top 20%	-0.62	1.88	3.84	11.04
7	Top 10%	-	2.60	4.19	10.75
8	Top 5%	1.54	3.12	4.22	9.64
9	Top 1%	2.88	3.27	4.39	8.34

Future Steps

- Fully implement the construction of $\hat{e}$ to estimate the equilibrium wage function and use it to identify d.

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- Use the full panel to separate d^t from d^s.
- Counterfactual analysis with increases in d—The new $100,000 H-1B fee.

Questions?

Thank You!

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General Separating Equilibrium

The general model follows the work of (Mailath, 1987).

Each worker chooses their education according to $\sigma : \Theta \to \mathcal{E}$.
Observing e, the firm updates belief about the worker's type $\widehat{\Theta}(e) \subseteq \Theta$.

In a full **separating** equilibrium, σ is a one-to-one function, i.e.,

$$\widehat{\Theta}(e) = \{\sigma^{-1}(e)\}, \forall e \in \mathcal{E} \tag{1}$$

and $\sigma(\theta)$ must satisfy *strict incentive compatibility*:

$$\{\sigma(\theta)\} = \underset{e}{argmax}\ U(\theta, \sigma^{-1}(e), e), \forall \theta \in \Theta \tag{2}$$

Regularity Conditions

1. (Smoothness) $U(\theta, \hat{\theta}, e)$ is twice differentiable on $\Theta^2 \times \mathcal{E}$.
2. (Belief Monotonicity) $\dfrac{\partial}{\partial \hat{\theta}} U\left(\theta, \hat{\theta}, e\right) > 0$.
3. (Type monotonicity) $\dfrac{\partial^2}{\partial \theta \partial e} U\left(\theta, \hat{\theta}, e\right) > 0$
4. ("Strict" Quasi-Concavity) $\frac{\partial}{\partial e} U\left(\theta, \hat{\theta}, e\right) = 0$ has a unique solution in e, denoted $\phi(\theta)$ such that $\phi(\theta) \in \underset{e}{argmax}\ U\left(\theta, \hat{\theta}, e\right)$, and $\dfrac{\partial^2}{\partial e^2} U\left(\theta, \hat{\theta}, e\right)\Big|_{e=\phi(\theta)} < 0$.
5. (Boundedness) $\exists k > 0$ such that

$$\forall (\theta, e) \in \Theta \times \mathcal{E}, \frac{\partial^2}{\partial e^2} U(\theta, \hat{\theta}, e) \geq 0 \ \Rightarrow \ \left| \frac{\partial}{\partial e} U(\theta, \hat{\theta}, e) \right| > k.$$

Additionally, there are two niceness conditions: Initial Value (IV) and Single Crossing (SC).

Characterization of the Solution

Mailath (1987) shows that
If conditions 1-5 are satisfied, then a strictly increasing function σ, continuous at $\underline{\theta}$, satisfies strict incentive compatibility if and only if:

1. σ solves $\dfrac{d\sigma}{d\theta} = -\dfrac{\frac{\partial}{\partial \hat{\theta}} U\left(\theta, \hat{\theta}, e\right)}{\frac{\partial}{\partial e} U\left(\theta, \hat{\theta}, e\right)}$
2. when $\frac{\partial}{\partial \hat{\theta}} U\left(\theta, \hat{\theta}, e\right) > 0$, $\dfrac{\frac{\partial}{\partial e} U\left(\theta, \hat{\theta}, e\right)}{\frac{\partial}{\partial \hat{\theta}} U\left(\theta, \hat{\theta}, e\right)}$ is a strictly increasing function of θ.

Moreover, the strictly increasing σ that satisfies strict incentive compatibility is unique.

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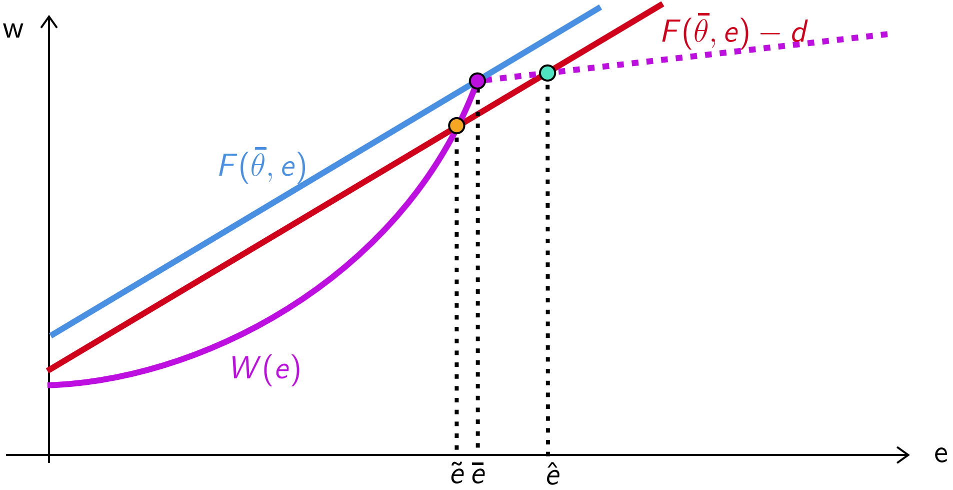
Moreover, the strictly increasing σ that satisfies strict incentive compatibility is unique.

In other words, there is an unique full separating equilibrium in the natives' labor market given the assumptions in the basic model environment.

New Separating Equilibrium



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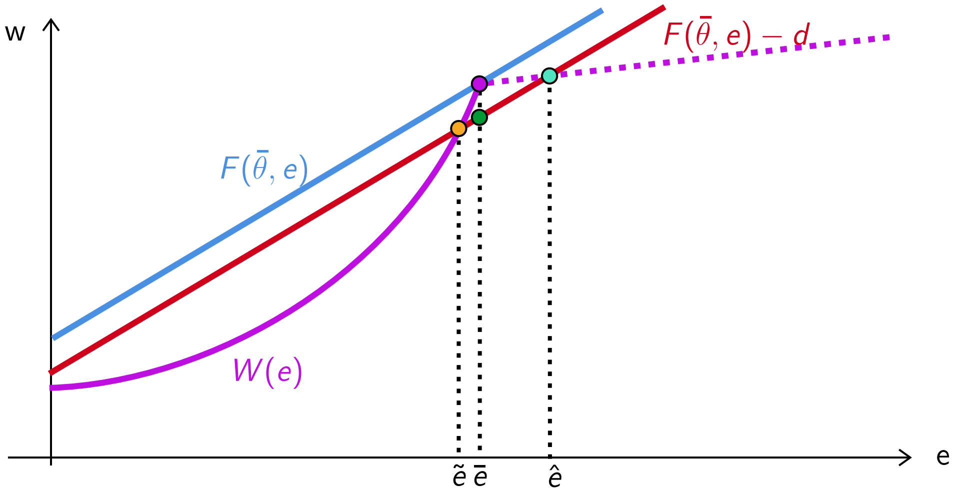

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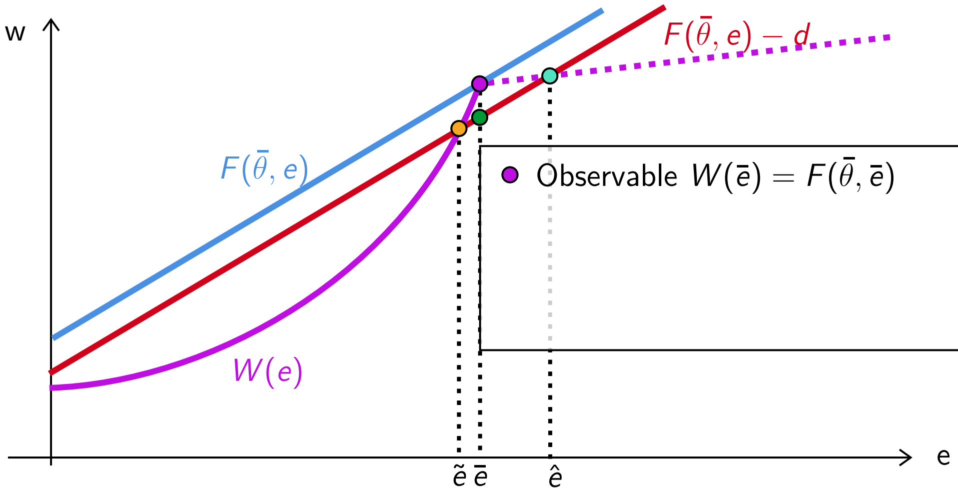

- Observable $W(\bar{e}) = F(\bar{\theta}, \bar{e})$

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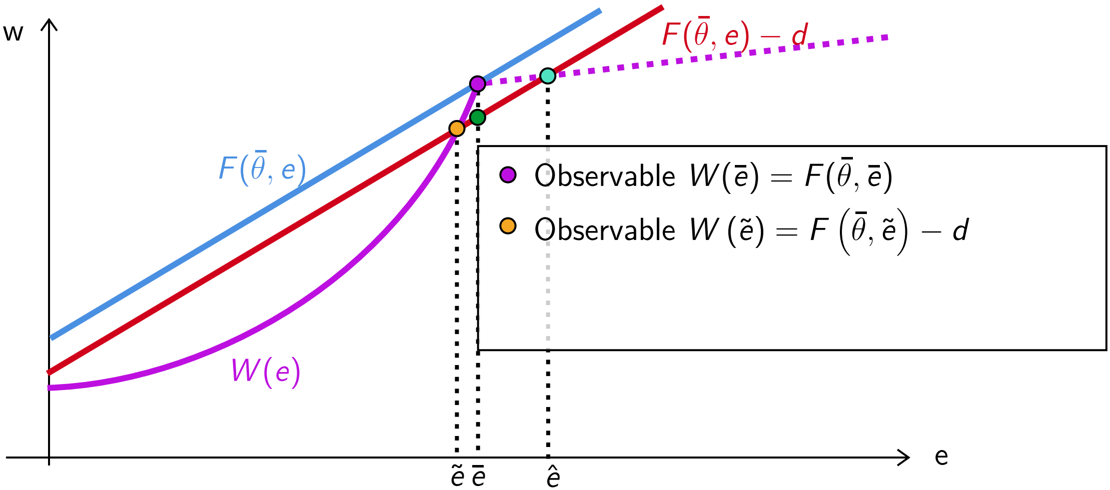

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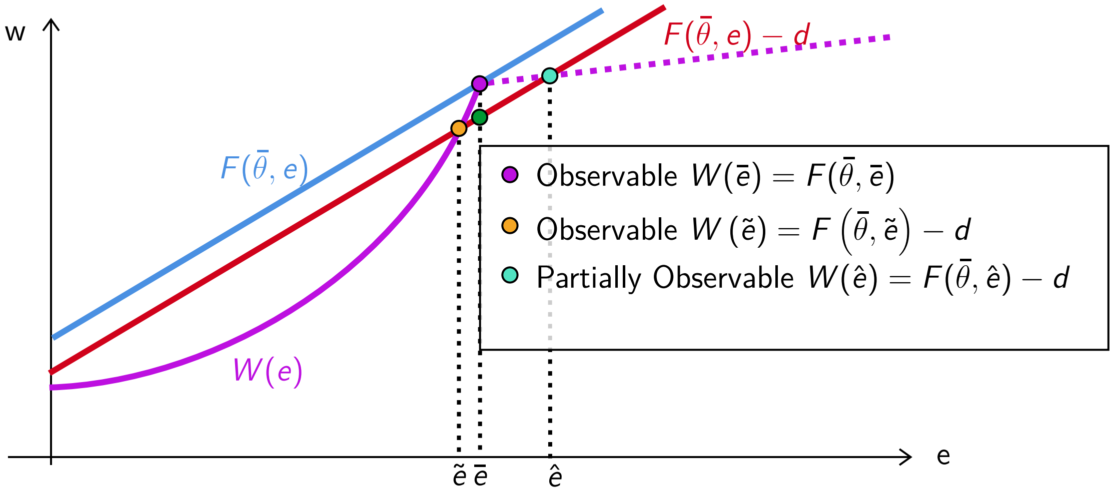

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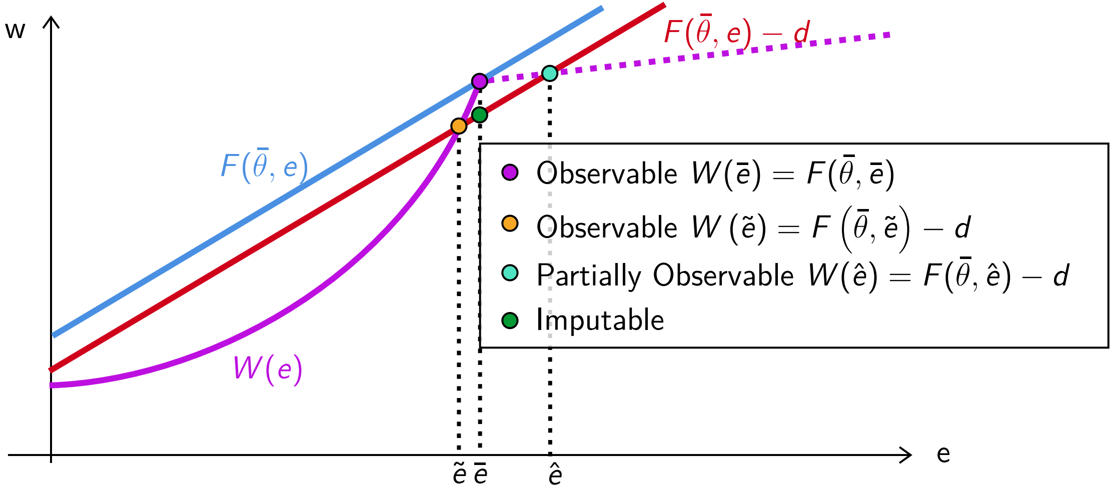
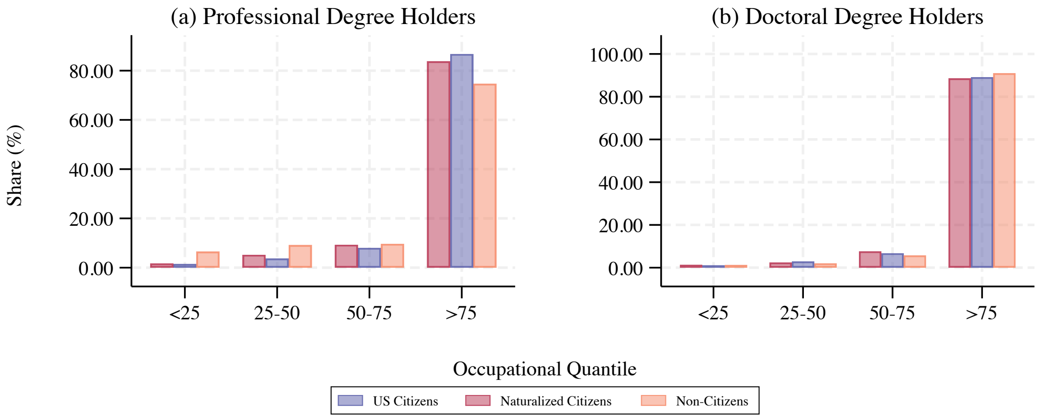


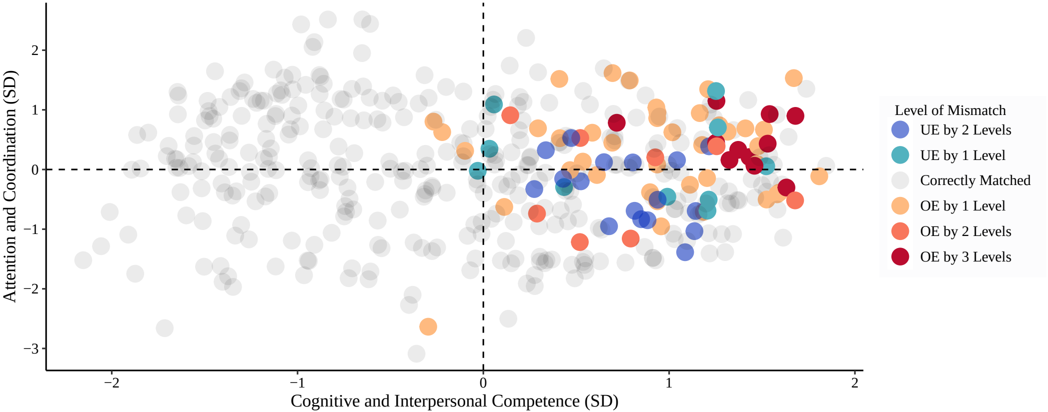
Figure A2: Educational Mismatch of Single Immigrants at High Degree Levels, 2010-2019



Notes: Data from American Community Survey-5% from 2010 to 2019. Occupational quantile represents the lexicographic order of jobs over the shares of U.S. citizen workers' education attainment of Doctorate/Professional, Master's, College, Associate's, and High School.

Back

Figure A3: Educational Mismatch of Non-Citizens, Highest Degree Obtained in US



Notes: Level of mismatch is the difference between the modal degree of a non−citizen worker, whose highest degree is obtained inside the United States, and the modal degree of a citizen worker. The degree levels, from highest to lowest, are: Doctorate, Professional, Master's, Bachelor's, Associates, High School, and less than High School.

Back

Figure A4: Educational Mismatch of Non-Citizens, Highest Degree Obtained in US

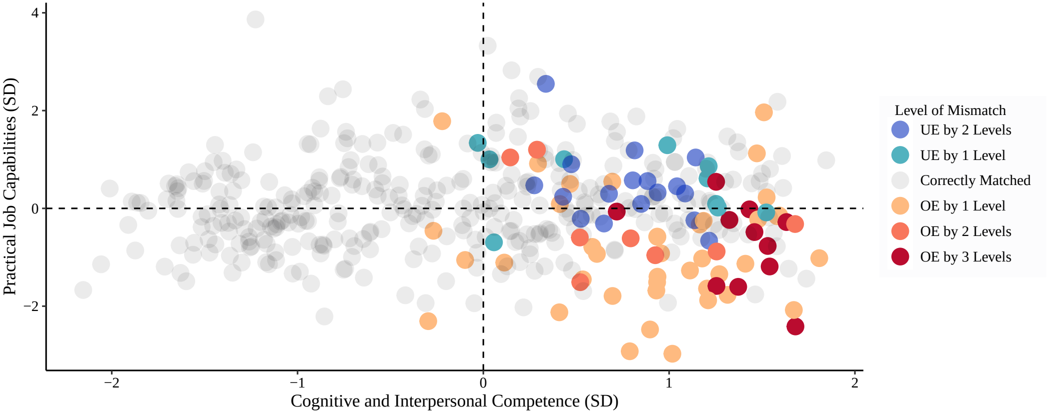


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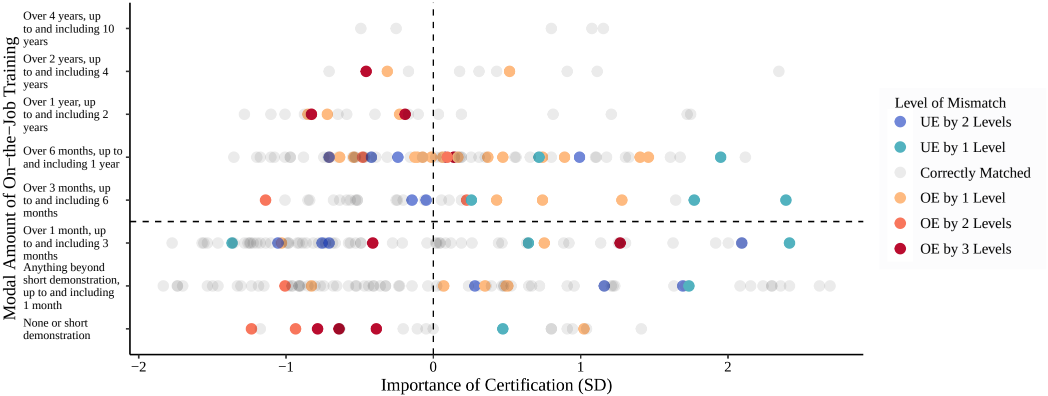
Figure A5: Educational Mismatch of Non-Citizens, Highest Degree Obtained in US



Notes: Level of mismatch is the difference between the modal degree of a non−citizen worker, whose highest degree is obtained inside the United States, and the modal degree of a citizen worker. The degree levels, from highest to lowest, are: Doctorate, Professional, Master's, Bachelor's, Associates, High School, and less than High School. Job characteristics are from the O*NET dataset. Mismatch realizations are from the American Community Survey 2010−2019.

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Graphical Representation

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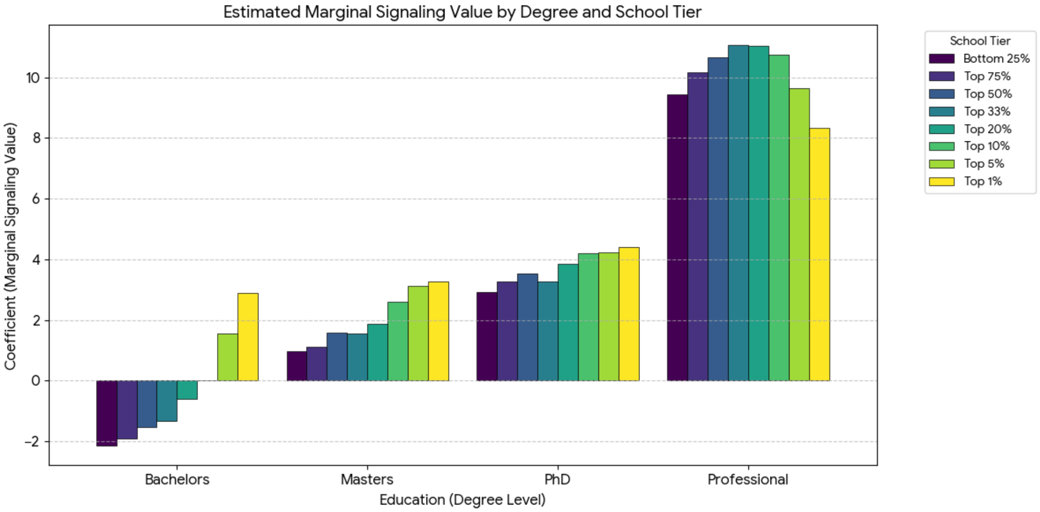

Location of Highest Degree

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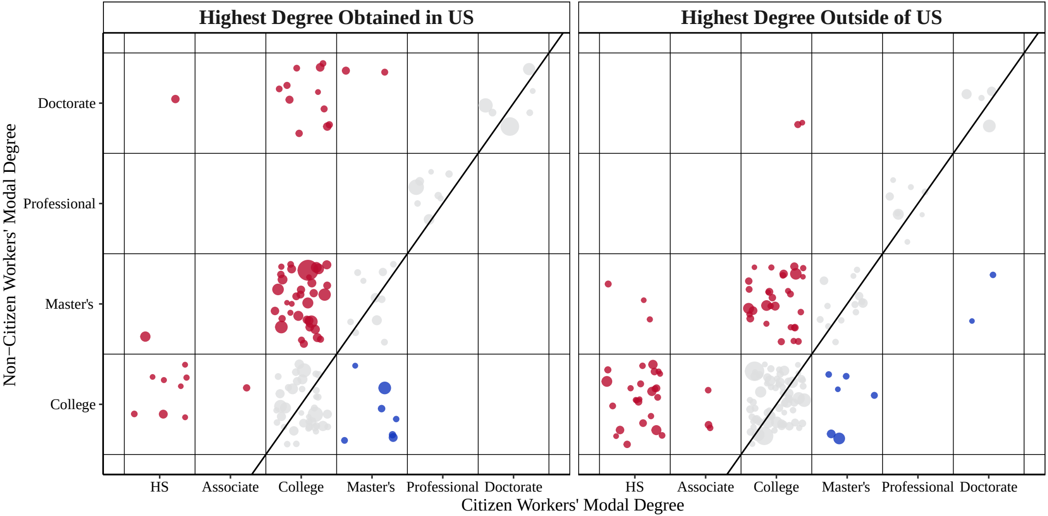
Figure A6: Educational Mismatch of Non-Citizens, 2010-2019



Location of Highest Degree

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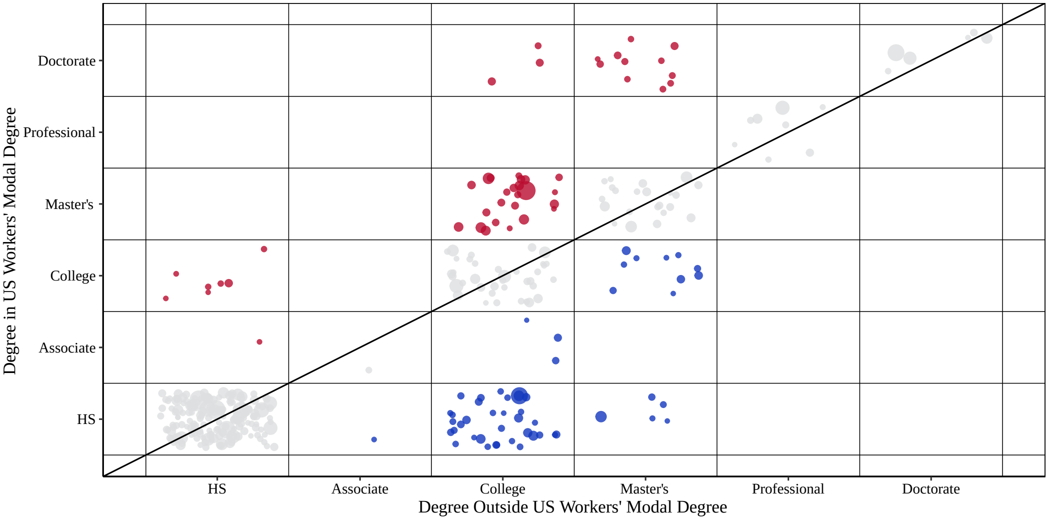
Figure A7: Educational Mismatch of Non-Citizens, 2010-2019



Location of Highest Degree

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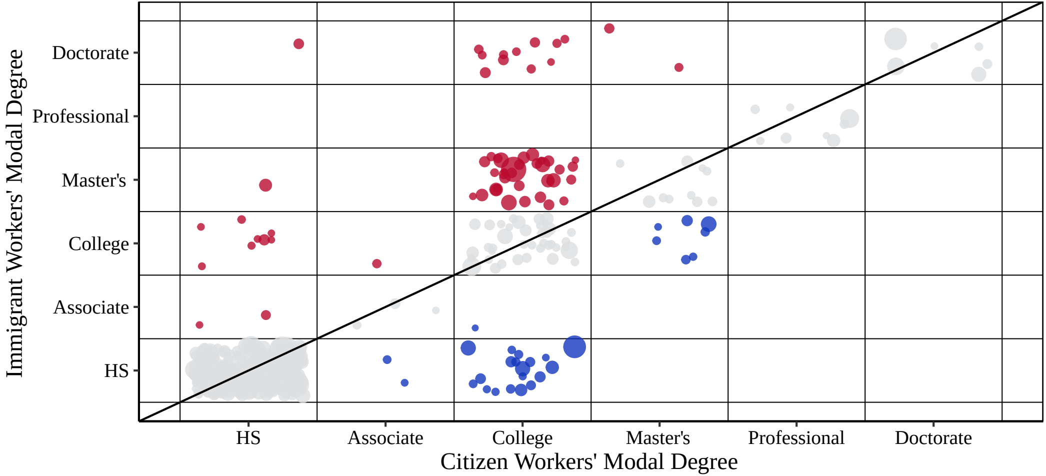
Figure A8: Educational Mismatch of Non-Citizens, Highest Degree in U.S., 2010-2019



Location of Highest Degree Back

Figure A9: Educational Mismatch of Non-Citizens, Highest Degree Outside of U.S., 2010-2019